pfi

Python for Fixed Income

COURSE 5B · TIME SERIES MODELS

Exponential Smoothing

WEEK 2 · DAY 1

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Today: six steps

Six steps

1. Build the EWMA by hand, in one formula
2. Fit simple smoothing: alpha by least squares
3. Add Holt's trend and a damped trend
4. Read every setting, sweep the ones that matter
5. Score walk-forward against the last value
6. EWMA of squared changes: a volatility description

The question

Can a smoother be closer to the next level than the last value is? Alpha equal to 1 is the last value, so: does least squares ever move away from 1? Development years only: 2 Jan 2015 to 30 Dec 2022, 2,001 dates.

Smoothing is scored on levels, so the last value scores day 2's table exactly.

Names for a model's output

A smoothing's output is a one-step value for a date. The fitted change is that value less the last value. The baseline:

Allowed names

fitted change in bp

the model's one-step value

the last value carried forward

Never called

a forecast or a prediction

a projection or an outlook

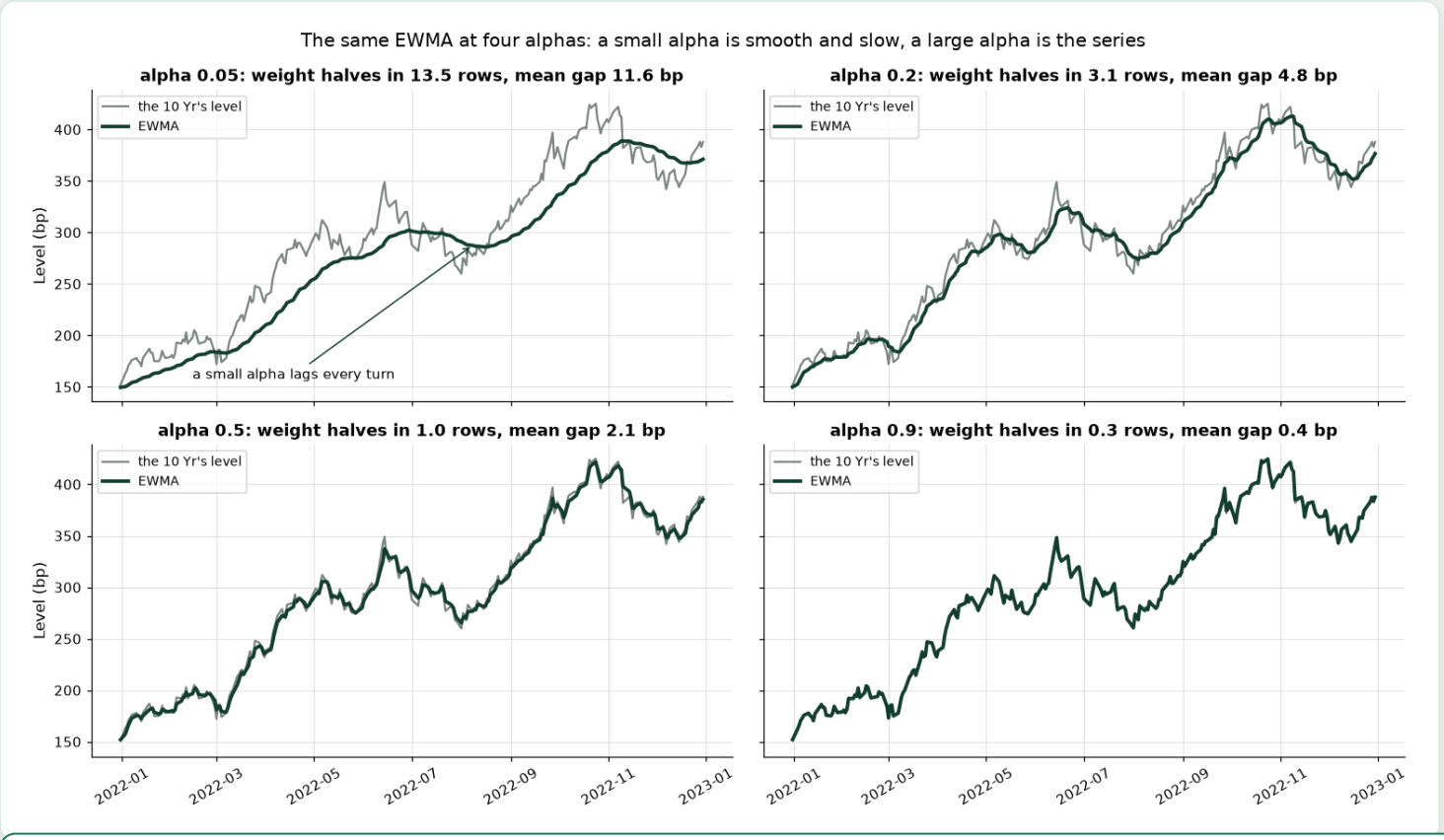
a signal, a call, a view on rates

Every score is on dates already in the file, from a model fitted on earlier dates.

How the EWMA works

The problem	A moving average drops a row all at once; an EWMA lets older rows fade
The recursion	New average = alpha x new value + (1 - alpha) x the average before
Alpha	Between 0 and 1: small is slow and smooth, 1 returns the series itself
What comes out	A smooth series on the same dates
The catch	The average through a date includes that date: shift it one row to use it as a value for the date

Alpha in four panels

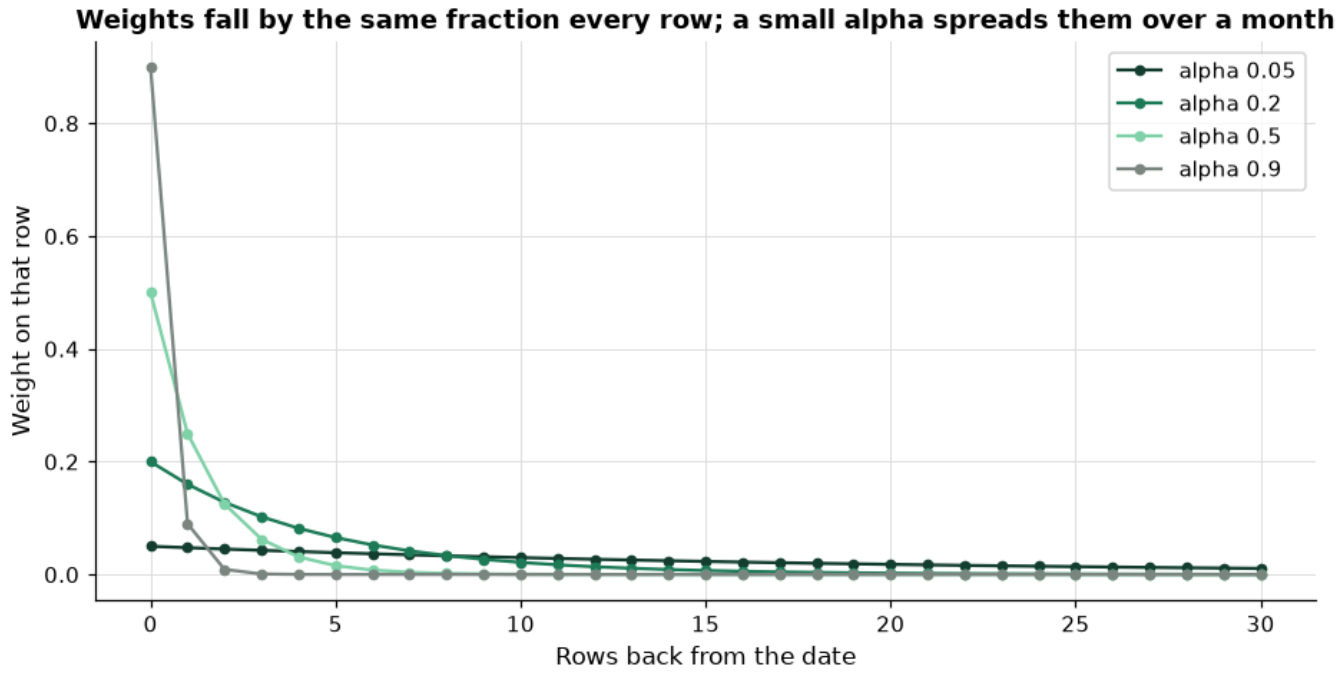


alpha	Halves (rows)	Gap (bp)
0.05	13.5	11.6
0.2	3.1	4.8
0.5	1.0	2.1
0.9	0.3	0.4

A small alpha is smooth and trails every turn. A large alpha is the series. Pick alpha by what the smoothing is for.

The EWMA is one recursive formula, copied down a column in Excel.

The weights



How to read it

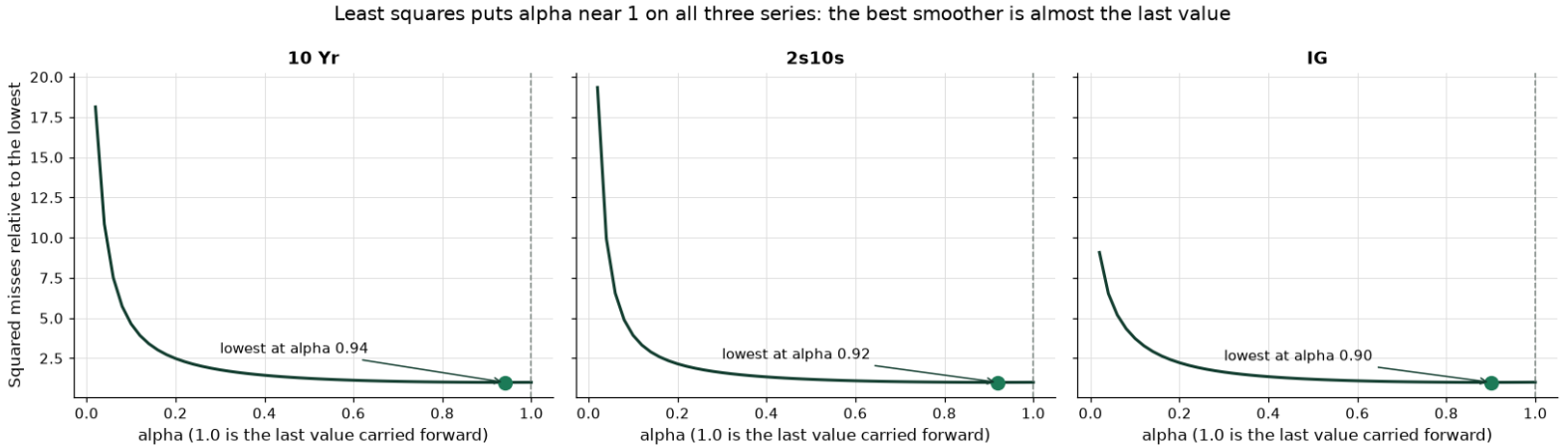
Weights fall by the same fraction every row. Alpha 0.05 still gives a month of history a weight; alpha 0.5 keeps almost nothing beyond 5 rows back.

Weight on the value k rows back: alpha x (1 - alpha) to the power k.

How simple smoothing works

The model	The next level is the average so far, plus a shock
The one-step value	The EWMA through the row before
The fit	Alpha makes the sum of squared one-step misses smallest; the starting level is fitted too
What comes out	Alpha, the starting level, and the sum of squared misses
The question	Alpha equal to 1 is the last value: how much averaging do the data want?

Alpha by least squares



Series	alpha	Starting level (bp)	First value (bp)
10 Yr	0.9426	211.5175	212.0000
2s10s	0.9173	145.1471	146.0000
IG	0.8947	90.2570	90.3000

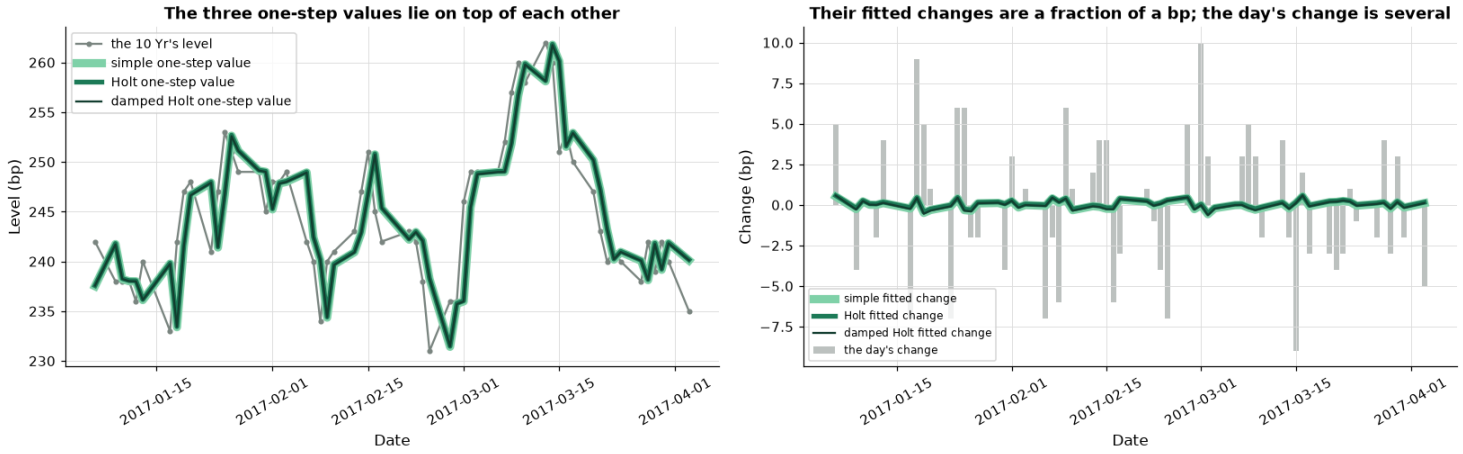
With the starting level fixed at the first value, statsmodels' one-step values are the EWMA of the row before (alpha 0.9430).

Simple smoothing and the EWMA are one model; the fit is only the choice of alpha.

How Holt's trend works

Two states	A level and a trend: the average change per row
The update	Level from the old level plus trend; trend as a weighted average (beta) of the latest level change and the old trend
The value	Level plus trend
Damped	The trend is multiplied by phi below 1 each row, which fades it
What comes out	Alpha, beta, phi, the starting values, and AIC and BIC

Three smoothers on the 10 Yr

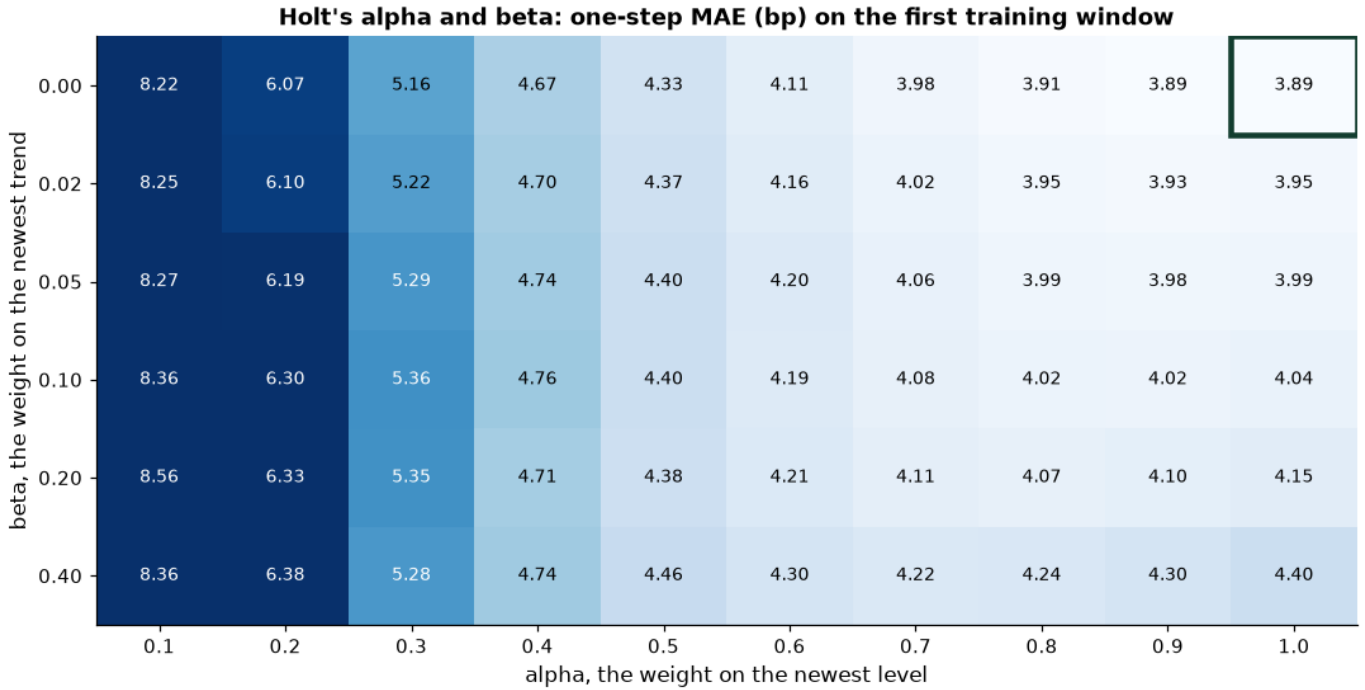


Model	alpha	beta	phi	AIC	BIC
simple	0.9426	-	-	1613.2229	1621.6680
Holt	0.9423	0.0000	-	1617.1608	1634.0511
damped Holt	0.9361	0.0000	0.8000	1614.7660	1635.8789

Simple smoothing has the lowest AIC. Holt's beta is 0: the trend never updates. The damped phi sits at its bound.

The fitted changes are a fraction of a bp; the day's change is several.

Holt's alpha and beta



How to read it

The lowest cell is alpha 1.0, beta 0.00: 3.8869 bp, which is the last value carried forward (3.8869 bp). No smaller alpha and no trend does better, even in-sample.

In-sample, first training window. The score that counts is walk-forward.

Every setting of ExponentialSmoothing

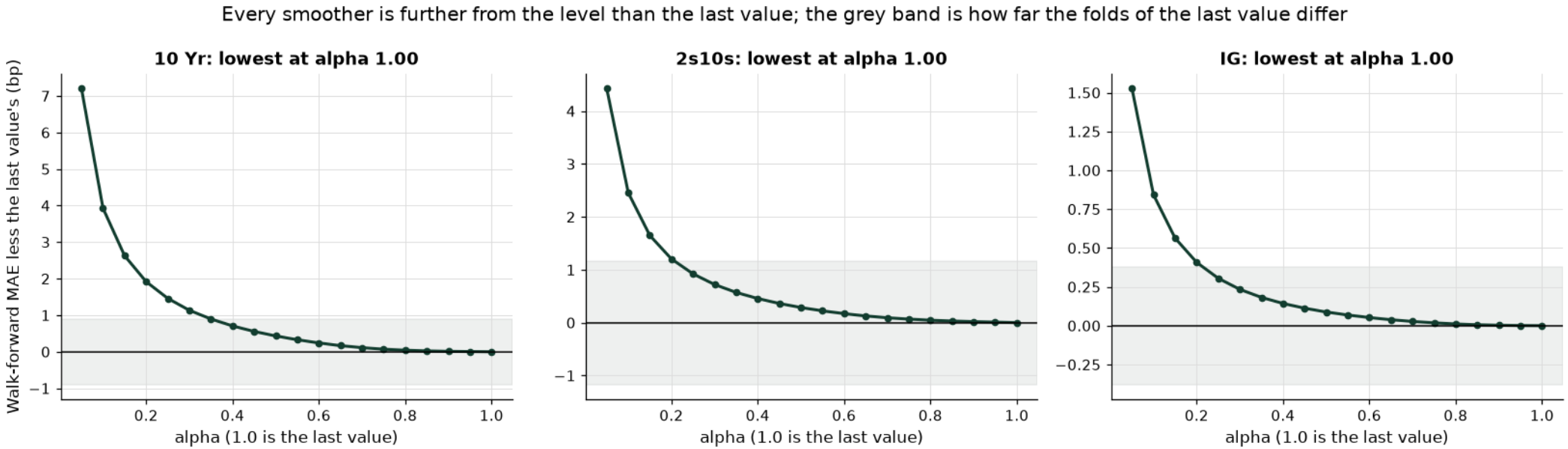
Setting	Default	Tune first
trend	None	yes
damped_trend	False	yes
initialization_method	'estimated'	yes
bounds	None	no
smoothing_level (fit)	None	yes
optimized (fit)	True	no

initialization_method	alpha	Start (bp)
estimated	0.9426	211.5175
heuristic	0.9429	211.8667
legacy-heuristic	0.9430	212.0000
known	0.9430	212.0000

Four ways to set the starting level give nearly the same alpha: its weight shrinks by (1 - alpha) every row.

The defaults are read from the installed statsmodels with inspect.signature, never typed.

A fixed alpha as a rule



Each dot is a fixed alpha scored walk-forward on day 2's folds. The curves fall to zero at alpha 1.0: the best fixed alpha on every series is the last value.

Every smoother below alpha 1 is further from the level than the last value.

How walk-forward smoothing works

Each fold	Fit the weights and the starting state once, on the training rows only
The test rows	Run the recursion through the training rows and on, with the weights held fixed
The value	The one-step value for a row uses only the rows before it
The next fold	Fitted again on a longer window
The proof	From day 2: the values must not move when the rows from the middle of the fold on are removed

Five rules against the last value

10 Yr	Gap (bp)	MASE	Below
drift	0.0076	1.0024	0
simple	0.0059	1.0019	3
Holt	0.0088	1.0028	1
damped	0.0066	1.0021	3
AR(1)	0.0089	1.0028	1

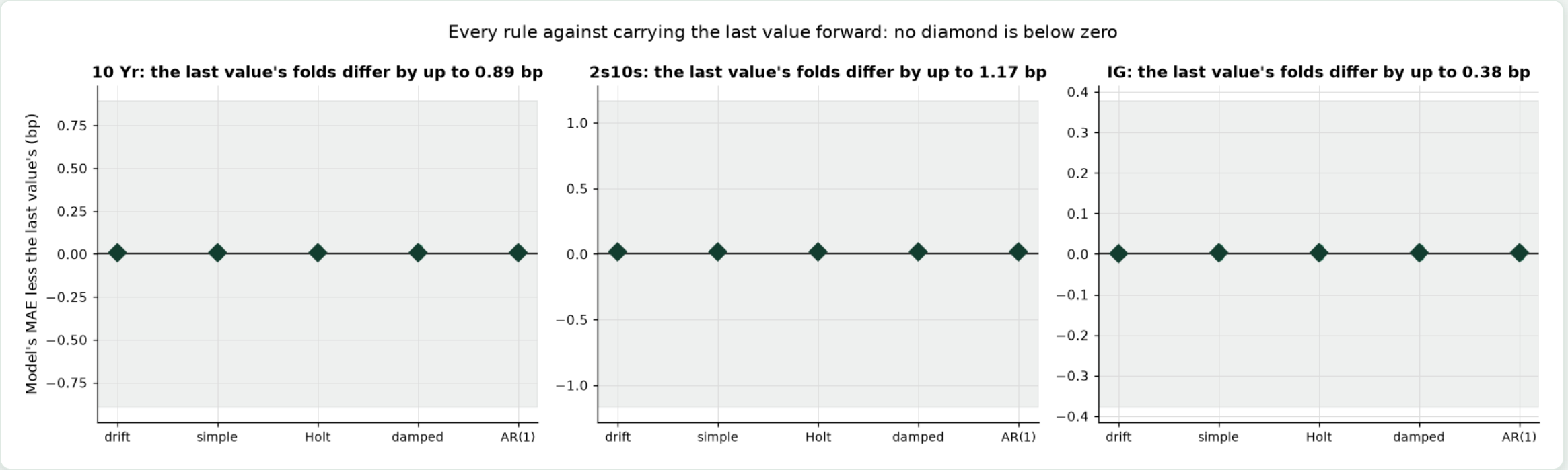
2s10s	Gap (bp)	MASE	Below
drift	0.0134	1.0062	0
simple	0.0135	1.0063	0
Holt	0.0155	1.0072	0
damped	0.0136	1.0063	0
AR(1)	0.0137	1.0064	0

IG	Gap (bp)	MASE	Below
drift	0.0003	1.0003	0
simple	0.0023	1.0028	2
Holt	0.0023	1.0029	2
damped	0.0023	1.0029	2
AR(1)	0.0024	1.0030	2

Gap: a rule's MAE less the last value's, five-fold average. Below: folds, out of 5, where the rule was below the last value. The last value's own folds differ by 0.8929 bp (10 Yr), 1.1667 (2s10s), and 0.3778 (IG).

Every MASE is above 1: no rule is closer than the last value on the five-fold average.

Every gap against its band



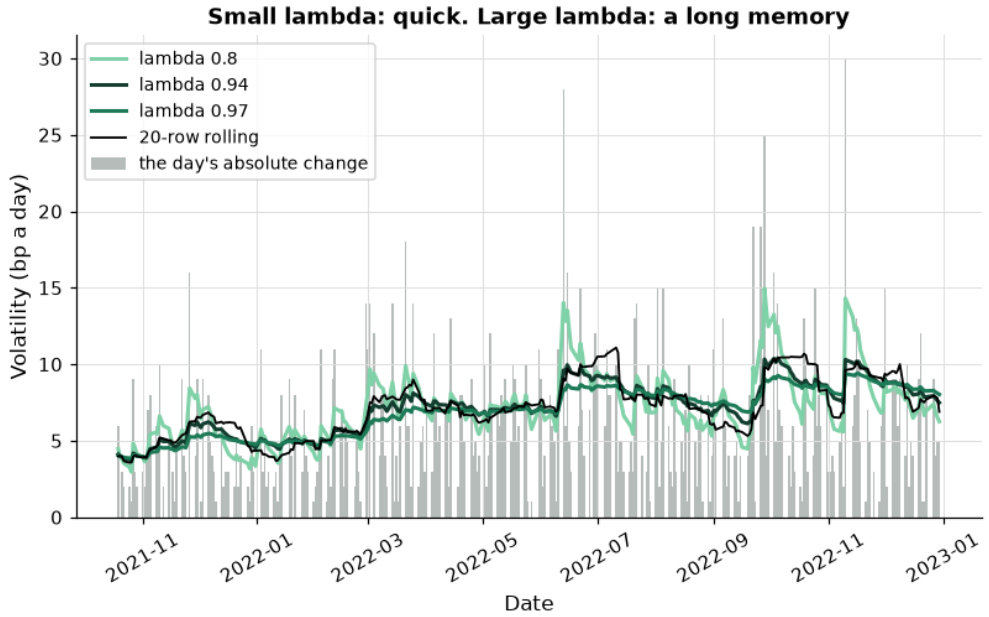
Every diamond sits a hair above zero, a tiny fraction of the band. A trend does not help: Holt's trend is no closer than simple smoothing.

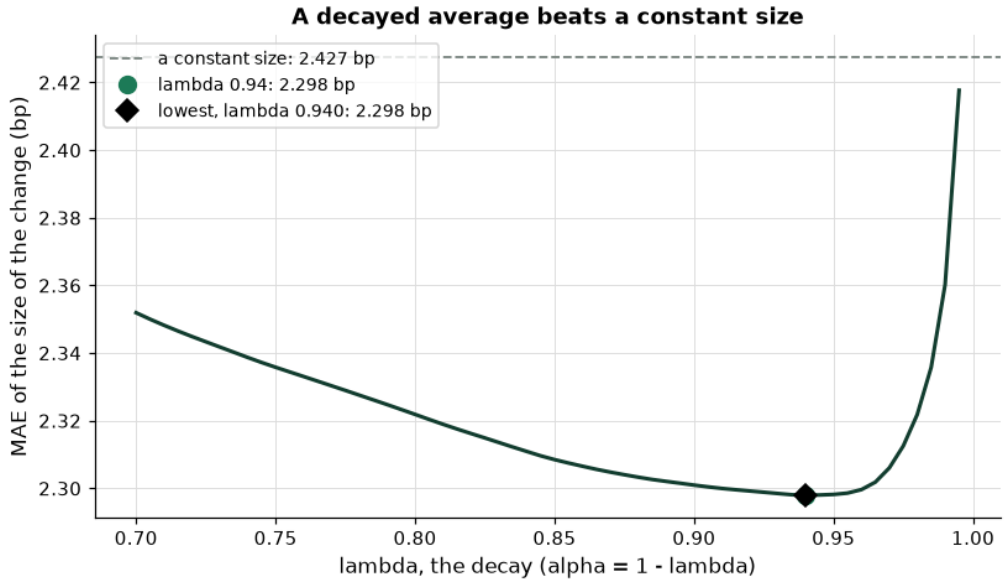
Dots are folds; diamonds are five-fold averages; the grey band is how far the last value's own folds differ.

How the RiskMetrics EWMA works

The idea	Squared changes measure how big the days are, and big days follow big days
The recursion	Variance now = lambda x variance before + (1 - lambda) x the latest change squared
Alpha	alpha = 1 - lambda; the square root is a volatility in bp a day
The choice	Lambda 0.94 for daily data, 0.97 for monthly: RiskMetrics Technical Document, 4th edition, December 1996
What comes out	A volatility for every date

Volatility at three decays





Lambda 0.94: 2.2979 bp against 2.4274 bp for a constant size. A description over the development years, not a score: day 7 scores it walk-forward. Volatility on 30 Dec 2022 is in the notebook.

Source: RiskMetrics Technical Document, 4th edition, J.P. Morgan and Reuters, December 1996 (published by MSCI).

Excel side by side

Excel side by side (tested in Excel)

```
=$H$1*B3+(1-$H$1)*C2
=LN(0.5)/LN(1-H1)
=SUMPRODUCT(ABS(B506:B757-G505:G756))/ROWS(B506:B757)
=SUMPRODUCT(ABS(B506:B757-B505:B756))/ROWS(B506:B757)
=$H$2*E3+(1-$H$2)*D4^2
```

last date 376.5808
3.1063 rows
fold 1 MAE 2.7519
last value 2.7619
volatility 7.4891 bp

Fill the recursion down a column, and change H1: the whole column redraws.

Save the day: undoing a mistake

1. Commit

The code, then the three alphas

2. Mistake

Every alpha overwritten with 1.0

3. Look

git status --short, then git diff

4. Restore

git restore puts the committed file back

```
Day 6: least-squares alphas
Day 6: tskit with ewma, smoothing_values, fit_smoothing, and walk_forward_smoothing
```

Commit a good result before an experiment, so the experiment has something to be undone to.

The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.

1

Docstring

Which dates fit, which dates score, what may be filled: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Assert what the docstring promises. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: the average is used as the value for a date but includes that date's own level: bumping the level moves the value from 88.28 to 113.28.

Code that runs is not the same as code that is right.

What you can do now

- The EWMA as one recursive formula, in Python and in Excel
- Simple smoothing is the EWMA with a least-squares alpha
- Holt's trend and a damped trend: what each adds
- Every setting of ExponentialSmoothing, and the sweeps
- No smoother beat the last value; RiskMetrics' 0.94 describes size

Week 2, Day 2 Volatility clustering and GARCH